Name: Srinath
Phone: +1 (203)-864-1182
Email: srinathbrahmaroothu03@gmail.com

Senior Data Engineer

PROFESSIONAL SUMMARY:

- Data engineer with overall 9+ years of experience in big data technologies including Apache Spark, Kafka,Cassandra, HBase, and Hadoop Distribution with expertise in Spark components such as RDD, Spark Hadoop, Hive, and Spark Streaming.
- Excellent Experience in Designing, Developing, Documenting, Testing of ETL jobs and mappings in Server and Parallel jobs using Data Stage to populate tables in Data Warehouse and Data marts.
- Have experience in Apache Spark, Spark Streaming, Spark SQL, and NoSQL databases like HBase, Cassandra, and MongoDB.
- Establishes and executes the Data Quality Governance Framework, which includes end - to-end process and data quality framework for assessing decisions that ensure the suitability of data for its intended purpose.
- Experienced in migrating SQL databases to Azure technologies, such as Azure Data Lake, Azure Data Lake Analytics, and Azure SQL Database.
- Good experience in Amazon Web Service (AWS) concepts like EMR and EC2 Webservices which provides fast and efficient processing of Teradata Big Data Analytics.
- Experienced in developing enterprise-level solutions using Hadoop architecture components including YARN, HDFS, MapReduce, and Apache Spark.
- In-depth knowledge of parallel processing and distributed systems architecture, with expertise in MapReduce and Spark execution framework.
- Proficient in data ingestion, processing and optimization using tools such as Sqoop, PIG, Hive, HBase, Oozie, Flume, NiFi, Kafka, Zookeeper, and Spark Context, Spark-SQL, Dataframe API, Spark Streaming, MLlib, and Pair RDDs.
- Skilled in using Amazon Web Services (AWS), including EC2, Glue, S3, RDS, VPC, IAM, Elastic Load Balancing, Auto Scaling, CloudWatch, SNS, SES, SQS, Lambda, EMR, and other AWS services.
- Skilled in data ingestion from various sources using Sqoop, Flume and transforming the data using Hive, MapReduce and loading it into HDFS.
- Adept in designing and implementing optimal performance tables in Hive using partitions and bucketing and familiar with different file formats such as Avro, parquet, ORC, JSON, and XML.
- Experience in developing Python, Scala, PySpark and SQL scripts in Databricks and EMR platforms to perform various transformations on the source data and get required output as per business needs.
- Solid experience developing PySpark Applications for performing highly scalable data transformations using RDD, Data frame, Spark-SQL, and Spark Streaming.
- Experienced in handling database issues and connecting with SQL and NoSQL databases, such as MongoDB, HBase, Cassandra, SQL Server, and PostgreSQL.
- Expert in creating Python scripts for handling data in MongoDB and HBase and using Phoenix to create a SQL layer on HBase.
- Used Informatica Power Center for (ETL) extraction, transformation and loading data from heterogeneous source systems into target database.
- Experienced in writing, testing and running the code and algorithms on the data to make sure they run and work as expected.

TECHNICAL SKILLS:

Big Data Technologies	Hadoop, MapReduce, Spark, HDFS, Sqoop, YARN, Oozie, Hive, Impala, Zookeeper, Apache Flume, Apache Airflow, Cloudera, HBase
Programming Languages	Python, PL/SQL, SQL, Scala, C, C#, C++, T-SQL, Power Shell Scripting, JavaScript
Cloud Services	Azure Data Lake Storage Gen 2, Azure Data Factory, Blob storage, Azure SQL DB, Databricks, Azure Event Hubs, AWS RDS, Amazon SQS, Amazon S3, AWS EMR, Lambda, AWS SNS
Databases	MySQL, SQL Server, Oracle, MS Access, Teradata, and Snowflake
NoSQL Data Bases	MongoDB, Cassandra DB, HBase
Development Strategies	Agile, Lean Agile, Pair Programming, Waterfall and Test-Driven Development
Visualization & ETL tools	Tableau, Informatica, Talend, SSIS, and SSRS
Version Control	Jenkins, Git, and SVN
Operating Systems	Unix, Linux, Windows, Mac OS
Monitoring tool	Apache Airflow

PROFESSIONAL EXPERIENCE:

Citizens Financial Group - Providence, Rhode Island **June 2024 to Present**
Senior Data Engineer

Responsibilities:

- Azure Data Factory was used to design, create, and maintain data pipelines for effective data extraction, transformation, and loading (ETL) procedures.
- Automated data ingestion from diverse sources (APIs, flat files, relational databases), ensuring daily updates of critical datasets.
- Integrated various data sources, including on-premises and cloud-based databases, into a unified data platform on Azure, ensuring data consistency and accuracy.
- Built and maintained scalable batch data pipelines using Apache Airflow, increasing system reliability and maintainability.
- Leveraged Azure Databricks for scalable and high-performance data processing, implementing data transformations and machine learning workflows.
- Developed and optimized complex SQL queries and stored procedures for data transformation and reporting tasks.
- Managed Azure SQL Database instances, optimizing performance, ensuring data security, and implementing disaster recovery strategies.
- Maintained and administered PostgreSQL and MySQL databases, including indexing, performance tuning, and schema migrations.
- Designed and implemented data models using Azure Synapse Analytics (formerly SQL Data Warehouse) to support analytical and reporting requirements.
- Processed and transformed multi-terabyte datasets using Apache Spark and PySpark for analytics and machine learning pipelines.
- Utilized Azure HDInsight to process and analyze large-scale data sets with technologies like Hadoop, Spark, and Hive.
- Involved in the construction of Web Services utilizing SOAP for sending and obtaining data from the external interface in the XML format.
- Load Salesforce CRM data into Azure Data Lake, Azure Synapse, and Snowflake for advanced analytics.
- Created portable and scalable data processing environments using containerization technologies like Docker or Kubernetes, allowing for easy deployment across many clusters and efficient resource use.
- Developed and tested disaster recovery strategies for Azure data solutions, ensuring data availability in the event of errors or outages.
- Knowledge of using Power Query to import and alter data from a variety of sources, including databases, spreadsheets, and web services.
- Utilizing programs like Apache JMeter or TPC-H, performance testing and benchmarking of data processing tasks were carried out with the goal of locating bottlenecks and pinpointing potential areas for optimization to improve scalability.
- Storing unprocessed data in optimized forms like ORC and Parquet in Azure HDInsight, Azure Blobs, and Azure tables.
- For simple Hadoop transitions, Spark applications were created using Python (Pyspark), and analytics on data in Hive were performed using Spark APIs via Hortonworks Hadoop YARN.

Environment: Azure, Azure Data Factory, Azure Data Lake Store (ADLS), Azure SQL, Blob storage, Azure SQL Data Warehouse, Azure HDInsight, Azure Databricks, Hive, Sqoop, Python, Spark SQL, Spark Streaming, Kafka, Shell Scripts, ORC, Parquet, Oozie, Cron Tab Schedulers.

Edward Jones - St.Louis, Missouri **Aug 2022 to May 2024**
Senior Data Engineer

Responsibilities:

- Used Azure Data Factory as an orchestration tool for integrating data from upstream to downstream systems. Automated jobs using different triggers (Event, Scheduled and Tumbling) in ADF.
- Involved in daily production server check list, SQL Backups, Disk Space, Job Failures, System Checks, checking performance statistics for all servers using monitoring tool and research and resolve any issues, checking connectivity.
- Analyzed the data flow from different sources to target to provide the corresponding design Architecture in Azure environment.
- Established a formal EDM, MDM (Master Data Management) program that creates effective engagement between Business operation, EDM, delivery team and IT.
- Designed and developed SSIS (ETL) packages to validate, extract, transform and load data from OLTP system to the Data warehouse.
- Created Application Interface Document for the downstream to create new interface to transfer and receive the files through Azure Data Share.
- Designed and implemented Tables, Functions, Stored Procedures and Triggers in SQL Server 2016 and wrote the SQL.
- Perform ongoing monitoring, automation and refinement of data engineering solutions prepare complex SQL views, stored procedures in Azure SQL Datawarehouse and Hyperscale.
- Integrated Azure Active Directory authentication to every Cosmos DB request sent and demoed feature to Stakeholders.
- Created Build definition and Release definition for Continuous Integration (CI) and Continuous Deployment (CD).
- An open-source storage layer called Delta Lake, which is an extension of Azure Databricks, enables ACID transactions for Apache Spark and big data workloads.
- Created Spark applications with Pyspark and Spark-SQL for data extraction, transformation, and aggregation from various file formats for analysis and transformation of the data to reveal insights into customer usage patterns.

- Implemented data storage solutions using Azure Data Lake Storage, ensuring data organization, access control, and compliance with data governance policies.
- Conducted quarterly Data owner meetings to communicate upcoming Data Governance initiatives, processes, policies and best practices.

Environment: Azure Cloud, Azure Data Factory (ADF v2), Azure functions Apps, Azure Data Lake, BLOB Storage, Visual Studio 2012/2016, Microsoft SQL Server 2012/2016, SSIS 2012/2016, Teradata Utilities, Windows remote desktop, UNIX Shell Scripting, AZURE PowerShell, Data bricks, Python.

Cardinal Health - Dublin, Ohio **March 2020 to July 2022**
Data Engineer

Responsibilities:

- Design and construct new processes for data modeling and production, utilizing prototypes, algorithms, predictive models, and custom analysis.
- Imported data from AWS S3 into Spark RDDs and executed transformations and actions to enable scalable data processing in a cloud environment.
- Continuously research analytics products to develop new or improved predictive modeling products and stay informed about evolving business practices.
- Analyzing the model needs to be implemented eg: a deep learning algorithm (NLP, CNN, lstm), or machine learning algorithm (random forest, XG boost), or reinforcement learning algorithm the model research.
- Focused on high availability and fault tolerance when designing and deploying Spark applications in AWS EMR, ensuring robust performance under varying workloads.
- Identify and adopt best practices in reporting and analysis, ensuring data integrity, design, analysis, validation, and documentation are maintained.
- Automated routine data extraction, transformation, and loading (ETL) tasks using AWS Lambda, reducing manual effort and ensuring seamless integration.
- Built and optimized NiFi dataflows for integrating legacy systems, traffic monitoring sensors, GIS data, and asset management systems into the TxDOT analytics platform.
- Prepare technical reports to interpret data, identify alternatives, and recommend data revisions to support informed decision-making.
- Managed the migration of Tableau data workflows to Snowflake integrated with AWS S3, enhancing scalability and streamlining reporting processes.
- Simplified complex Tableau workflows by converting them into efficient single SQL queries, reducing dependency on Tableau and improving data retrieval performance.
- Utilized AWS Data Lake and Amazon Redshift for scalable data storage and querying, enabling high-performance analytics across large datasets.
- Implemented data validation, enrichment, and routing logic in NiFi to standardize datasets before loading into TxDOT's reporting and business intelligence tools (Power BI / Tableau).
- Contributed to cost optimization by refining workflows, leveraging AWS Glue and S3 tiered storage, and automating data handling to reduce resource usage.
- Developed reusable framework to be leveraged for future migrations that automates ETL from RDBMS systems to the Data Lake utilizing Spark Data Sources and Hive data objects.
- Created specialized data monitoring and warning systems utilizing AWS CloudWatch or Apache Kafka Connect to proactively spot and resolve any pipeline or data quality issues.
- Developed Kibana Dashboards based on the Log stash data and Integrated different source and target systems into Elasticsearch for near real time log analysis of monitoring End to End transactions.
- Used Spark code and Spark SQL/Streaming to process data more quickly in the AWS environment during testing.

Environment: AWS services like Glue, S3, Redshift, Lambda, Cloud Watch, Cloud Formation, Teradata, Snowflake, GIS Data, AI/ML, API, Terraform, Data Brew, SQL, Tableau prep,spark.

Early Warning's - Scottsdale, Arizona **May 2018 to Feb 2020**
Data Engineer

Responsibilities:

- Responsible for maintaining quality reference data in source by performing operations such as cleaning, transformation and ensuring integrity in a relational environment by working closely with the stakeholders & solution architect.
- Designed and developed Security Framework to provide fine grained access to objects in AWS S3 using AWS Lambda, DynamoDB.
- Established a formal EDM, MDM (Master Data Management) program that creates effective engagement between Business operation, EDM, delivery team and IT.
- Experience in data ingestions techniques for batch and stream processing using AWS Batch, AWS Kinesis, AWS Data Pipeline
- Designed and developed SSIS (ETL) packages to validate, extract, transform and load data from OLTP system to the Data warehouse.
- Performed end- to-end Architecture & implementation assessment of various AWS services like Amazon EMR, Redshift, S3.

- Set up and worked on Kerberos authentication principals to establish secure network communication on cluster and testing of HDFS, Hive, Pig and MapReduce to access cluster for new users.
- Used AWS EMR to transform and move large amounts of data into and out of other AWS data stores and databases, such as Amazon Simple Storage Service (Amazon S3) and Amazon DynamoDB
- Developed Spark applications using Pyspark and Spark-SQL for data extraction, transformation, and aggregation from multiple file formats.
- Implemented AWS Step Functions to automate and orchestrate the Amazon SageMaker related tasks such as publishing data to S3, training ML model and deploying it for prediction.
- Import the data from different sources like HDFS/HBase into Spark RDD and perform computations using Pyspark to generate the output response.
- Integrated Apache Airflow with AWS to monitor multi-stage ML workflows with the tasks running on Amazon SageMaker.
- Creating Lambda functions with Boto3 to deregister unused AMIs in all application regions to reduce the cost for EC2 resources.
- Designed and deployed Teradata data extraction procedures utilizing software like AWS Glue or Apache Nifi, preserving the integrity and consistency of the data during the migration.
- Importing & exporting database using SQL Server Integrations Services (SSIS) and Data Transformation Services (DIS Packages).
- Focused on high availability, fault tolerance, and auto-scaling when designing, developing, and deploying numerous applications leveraging the AWS stack (EC2, S3, and EMR).
- Worked with Tableau for generating reports and created Tableau dashboards, pie charts, and heat maps according to the business requirements.

Environment: Python, Kafka, Flask, NumPy, Pandas, SQL, MySQL, Cassandra, AWS EMR, Spark, GIT, AWS Kinesis, AWS Redshift, AWS EC2, AWS S3, AWS SNS, AWS Lambda, AWS data pipeline, AWS cloud-watch, Airflow, Docker, Shell scripts, Snowflake.

Info Logitech Systems - Hyderabad, India **July 2016 to April 2018**
Data Analyst

Responsibilities:

- Performed exploratory data analysis using Python libraries such as Pandas and NumPy to identify patterns, trends, and anomalies in large datasets.
- Developed SQL queries to extract, transform, and aggregate data from multiple sources to support operational and strategic decision-making.
- Prepared presentation decks and reports to summarize analytical findings for both technical and non-technical audiences.
- Collaborated with cross-functional teams to gather business requirements and translate them into analytical solutions using SQL and Power BI.
- Prepared executive summaries and briefing documents based on deep dives into customer, sales, and product usage data.
- Documented SQL queries, Python scripts, and dashboard logic to improve knowledge sharing and onboarding processes.
- Partnered with stakeholders to define key performance indicators (KPIs) and built reporting frameworks aligned with business goals.
- Analyzed marketing campaign performance by segmenting audiences, tracking engagement, and mapping conversions.
- Developed automated Excel reports using pivot tables, VLOOKUPs, and macros to streamline weekly and monthly reporting tasks.
- Supported senior analysts in A/B testing, helping evaluate marketing campaign effectiveness and product feature adoption.
- Built custom Python scripts for data preprocessing, enabling faster turnaround time for ad hoc requests from business teams.
- Documented analytical processes and data dictionaries to ensure smooth knowledge transfer and report reproducibility.
- Collaborated with product managers and engineers to define and track user behavior metrics for a new feature rollout.
- Conducted cohort and retention analysis using Python to uncover user engagement trends and inform marketing strategies.
- Supported the product analytics team in tracking key metrics like DAU, MAU, and feature click-through rates using Mixpanel and Google Analytics.
- Worked on a capstone project to identify churn predictors from subscription and user activity data, presenting findings to leadership.

Environment: Python (Pandas, NumPy, Matplotlib), SQL (PostgreSQL, MySQL), Power BI, Excel (Pivot Tables, VLOOKUP, Macros), Google Analytics, Mixpanel, Jupyter Notebooks, Git.